

STA1023/MAT1063

Introduction to Probability Theory

Chapter 1 Counting Techniques

Hemalika T.K. Abeyesundara, PhD.

Department of Statistics & Computer Science

University of Peradeniya

Chapter 1 – Counting Techniques

- Basic Principle of Counting
- Permutations
- Combinations

Introduction

In many problems of statistics, we must list all the alternatives that are possible in a given situation or at least determine how many different possibilities are there.

In connection with latter, we often use the following theorem called '**Basic Principle of Counting**'.

Theorem 1.1 : Basic Principle of Counting

If an operation consists of two steps, of which the first can be done in n_1 ways and for each of these the second can be done in n_2 ways, then the whole operation can be done in $n_1 \cdot n_2$ ways.

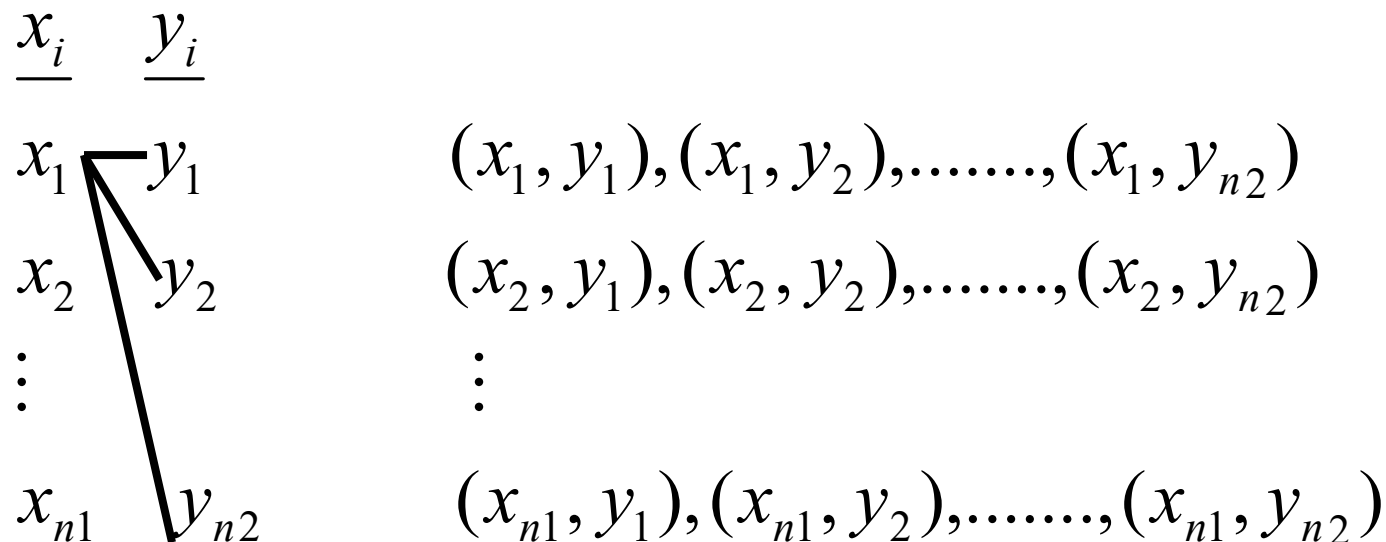
Here the operation stands for any of procedure, process or method of selection.

Justification of Theorem 1.1

Let us define an outcome of an operation with the ordered pair (x_i, y_i) .

Suppose the first step (x_i) results in probability n_1 and second step (y_i) results in probability n_2 .

Then the set of all possible outcomes is composed of $n_1 \cdot n_2$ pairs.

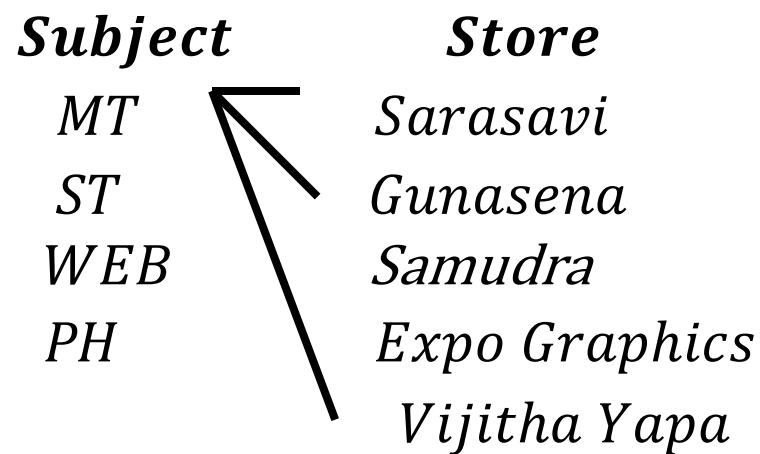


Example 1.1:

Suppose that a student has money to buy one book out of a Mathematics book, Statistics book, Web Designing book or a Physics book from one of the five stores Sarasavi, Gunasena, Samudra, Expo Graphics or Vijitha Yapa bookshop. Find the number of ways in which this can be done.

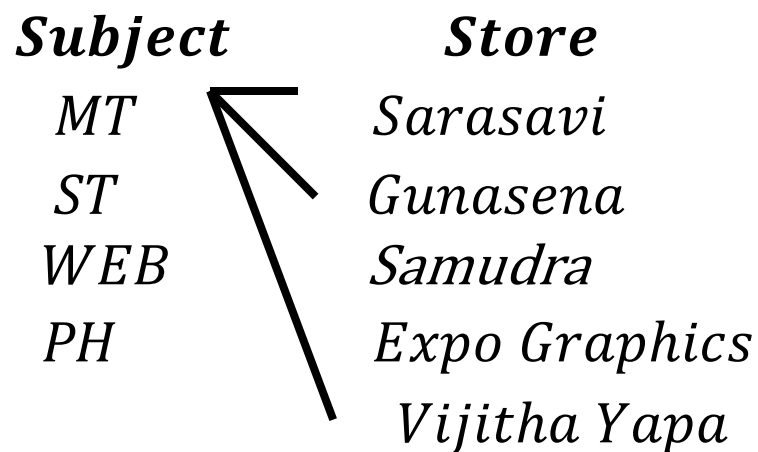
Example 1.1:

Suppose that a student has money to buy one book out of a Mathematics book, Statistics book, Web Designing book or a Physics book from one of the five stores Sarasavi, Gunasena, Samudra, Expo Graphics or Vijitha Yapa bookshop. Find the number of ways in which this can be done.



Example 1.1:

Suppose that a student has money to buy one book out of a Mathematics book, Statistics book, Web Designing book or a Physics book from one of the five stores Sarasavi, Gunasena, Samudra, Expo Graphics or Vijitha Yapa bookshop. Find the number of ways in which this can be done.



Number of ways to choose the subject= $n_1=4$

Number of ways to choose the bookshop= $n_2=5$

Total number of possible ways = $n_1 \cdot n_2 = 4 \cdot 5 = 20$

Example 1.2:

How many possible ways (outcomes) are there when we roll a pair of dice, one is red and the other one is green.

Example 1.2:

How many possible ways (outcomes) are there when we roll a pair of dice, one is red and the other one is green.

Number of outcomes from red dice= $n_1=6$

Number of outcomes from green dice= $n_2=6$

Total number of possible ways = $n_1 \cdot n_2 = 6 \cdot 6 = 36$

Theorem 1.1 can be extended to cover situations where an operation consists of more than two steps.

Theorem 1.2

If an operation consists of k steps, of which the first one can be done in n_1 ways, for each of these the second step can be done in n_2 ways, for each of the first two the third step can be done in n_3 ways, and so forth.

Then the whole operation can be done in

$$n_1 \cdot n_2 \cdot \dots \cdot n_k$$

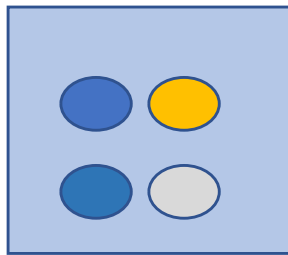
ways.

Example 1.3:

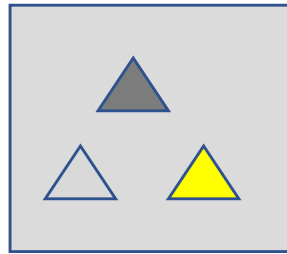
A quality control inspector wishes to select a part for inspection from each of four different bins containing 4, 3, 5 and 4 parts respectively. In how many different ways can the quality control inspector choose the four parts?

Example 1.3:

A quality control inspector wishes to select a part for inspection from each of four different bins containing 4, 3, 5 and 4 parts respectively. In how many different ways can the quality control inspector choose the four parts?

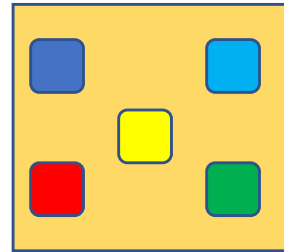


4



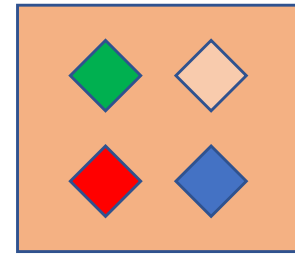
X

3



X

5



X

4

Total number of possible ways = **$4 \times 3 \times 5 \times 4 = 240$**

Example 1.4:

In how many different ways can a student answer all the questions of a true- false test containing 20 questions?

Number of ways to answer Q1 =2

Number of ways to answer Q2 =2

.

.

.

Number of ways to answer Q20 =2

Total number of possible ways = **$2 \times 2 \times 2 \times \dots = (2)^{20} = 1048576$**

Permutations

Definition 1.1

Different ways in which a group of objects can be ordered or arranged.

For example,

- i. How many different ways 5 books can be ranked as 1st, 2nd, 3rd.....
- ii. How many different ways 10 songs can be played in a play list.

Example 1.5:

How many permutations are there of the letters a, b and c.

A	B	C
A	C	B
B	A	C
B	C	A
C	A	B
C	B	A

Position
1

Position
2

Position
3

Number of letters available to fill the 1st position=3

Number of letters available to fill the 2nd position=2

Number of letters available to fill the 3rd position=1

Total number of arrangements = $3 \times 2 \times 1 = 6$

By generalizing the Example 1.5:

n distinct objects can be arranged in,

$$n * (n - 1) * (n - 2) * * 3 * 2 * 1$$

different ways.

Notation:

$$n * (n - 1) * (n - 2) * * 3 * 2 * 1 = n!$$

By definition:

$$\mathbf{0! = 1}$$

Theorem 1.3

The number of permutations of n distinct objects is $n!$.

Example 1.6:

In how many different ways can the first five players of a cricket team be sent to the ground?

Number of players available to go as the 1st = 5

Number of players available to go as the 2nd = 4

Number of players available to go as the 3rd = 3

Number of players available to go as the 4th = 2

Number of players available to go as the 5th = 1

Total number of arrangements = $5 \times 4 \times 3 \times 2 \times 1 = 5! = 120$

Example 1.7:

Find the number of ways (permutations) of 4 letters A, B, C and D.

Using Theorem 1.3, the answer is **$4! = 24$**

Example 1.7 modified:

Find the number of ways (permutations) of 4 letters A, B, C and D if two letters are selected among the 4

Example 1.7 modified:

Find the number of ways (permutations) of 4 letters A, B, C and D if two letters are selected among the 4

Now we have to fill only 2 positions....

Number of ways to fill the position1 =4

Number of ways to fill the position 2 = 3

Total number of arrangements = **4X3 =12**

A	B
A	C
A	D
B	A
B	C
B	D
C	A
C	B
C	D
D	A
D	B
D	C

Generalizing the above argument,

We find that n distinct objects taken r at a time for $r > 0$, can be arranged in,

$$n * (n - 1) * (n - 2) *(n - r + 1)$$

We denote this product by ${}^n P_r$

By definition, ${}^n P_0 = 1$

Theorem 1.4

The number of permutations of n distinct objects taken r at a time is

$${}_n P_r = \frac{n!}{(n-r)!} \quad ; \text{ for } n=0,1,2,\dots,n$$

Example 1.9:

In how many ways can an organizer schedule three different meetings if there are five available time slots?

Example 1.9 revisited:

In how many ways can an organizer schedule three speakers for three different meetings if they are all available on any of five possible time slots?

Number of slots available = $n =$

Number of meetings to be scheduled = $r =$

Total number of ways to schedule three meetings =

We have to arrange the speakers too..... ?

Number of days available= $n = 5$

Number of days to be selected = $r = 3$

Total number of ways to select 3 days =

$${}^nP_r = \frac{5!}{(5-3)!} = \frac{5!}{2!} = \frac{5*4*3*2*1}{2*1} = 5*4*3 = 60$$

By Theorem 1.3, number of ways to arrange three speakers = $3!=6$

By Theorem 1.1, number of ways the meeting can be scheduled=

$$\mathbf{60*6 = 360}$$

We have been assumed until now n objects from which we select r objects and form permutations are all distinct.

What will happen if there are repeating objects?

Example 1.10

How many different permutations are there of the letters in the word “GOOD”?

Example 1.10

How many different permutations are there of the letters in the word “GOOD”?

If we distinguish for the moment between the two O's by labeling them O_1 and O_2 there are $4!=24$ different permutations of the symbols G, O_1 , O_2 and D.

However if we drop the subscripts, then GDO_1O_2 and GDO_2O_1 for instance, both yield GDOO. Hence each pair of permutations with subscripts yields but one arrangement without subscripts.

Then total number of arrangements of the letter
“GOOD” = $24/2 = 12$

Example 1.11:

In how many different ways can four copies of one text book and one copy of each of four other text books can be arranged on a shelf?

Number of ways to arrange 7 text books considering all of them are distinct =

Number of identical copies=

Total number of ways to arrange the Text books=

Example 1.11:

In how many different ways can four copies of one text book and one copy of each of four other text books can be arranged on a shelf?

Number of ways to arrange 7 text books considering all of them are distinct = $7! = 5040$

Number of identical copies = 4

Total number of ways to arrange the Text books = $7!/4!$

Generalizing the above argument,

Theorem 1.5

The number of permutations of n objects of which n_1 are of one kind, n_2 are of a second kind,, n_k of a n^{th} kind and $n_1 + n_2 + \dots + n_k = n$ is

$$\frac{n!}{n_1! n_2! \dots n_k!}$$

Example 1.12:

In how many ways can two novels by A, three novels by B and two novels by C be arranged side by side on a bookshelf if we do not distinguish between the titles of the books by the same author?

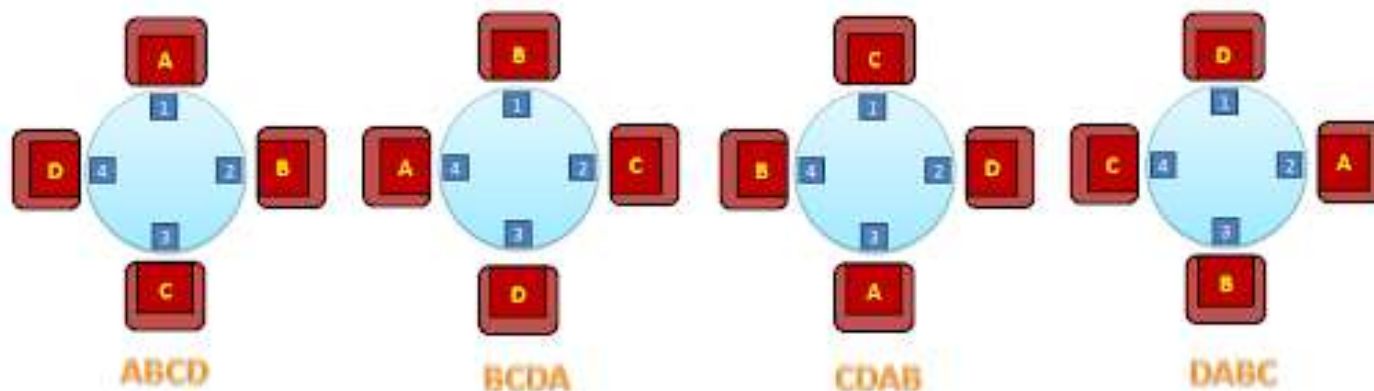
$$\frac{7!}{2!3!2!} = 210$$

Circular Permutations

Permutations that occur when objects are arranged in a circle are called **circular permutations**.

Two circular permutations are not considered different (and are count only once) if corresponding objects in the two arrangements have the same objects to their left and to their right.

For example, if four persons are playing cards, we do not get different permutation if everyone moves to his or her right.



Example 1.13:

How many circular permutations are there of four persons playing cards?

We have to fix one position and find all possible arrangements of the rest.

Then there are $(4-1)!$ Different arrangements = $3! = 6$

Generalizing the above argument,

Theorem 1.6

The number of permutations of n distinct objects arranged in a circle is

$$(n-1)!$$

Combinations

Determining the number of ways in which r objects can be selected from among n distinct objects **without regard to the order** in which they are selected.

Such selections are called combinations.



Example 1.14:

In how many distinct ways can a supervisor select three students among 20 students for a group meeting?

If we consider the order in which the students are selected, the number of permutations will be

$${}^{20}P_3 = 6840$$

But each of three students will then be counted $3! = 6$ times.

If we disregard the order, the total number of ways to select 3 out of 20 will be

$$\frac{{}^{20}P_3}{3!} = \frac{6840}{6} = 1140$$

In general there are $r!$ permutations of the object in a subset of r objects.

Since the order is considered, ${}^n P_r$ permutations of r objects selected from a set of n distinct objects contain each subset $r!$ times.

The number of combinations or number of subsets of r objects that can be selected from a set of n distinct objects can be obtained by dividing ${}^n P_r$ by $r!$.

The results is denoted by

$$\binom{n}{r} = {}^n C_r$$

Theorem 1.7

The number of combinations of n distinct objects taken r at a time is

$$\binom{n}{r} = {}^nC_r = \frac{n!}{r!(n-r)!}$$

for $r=0,1,2,\dots,n$

Example 1.15:

In how many different ways can two books be selected among six books?

$$\binom{6}{2} = {}^6C_2 = \frac{6!}{2!(6-2)!} = \frac{6!}{2!4!} = \frac{6*5*4!}{2!4!} = \frac{6*5}{2*1} = 15$$

Example 1.16:

How many different three member committees can be formed from a group of seven members where four are males and three are females?

$$\binom{7}{3} = {}^7C_3 = \frac{7!}{3!(7-3)!} = \frac{7!}{3!4!} = \frac{7 * 6 * 5 * 4!}{3!4!} = \frac{7 * 6 * 5}{3 * 2 * 1} = 35$$

Recalculate the number of committees if there should be two males and one female?

Number of ways to select two males out of 4 = ${}^4C_2 = \frac{4!}{2!(4-2)!} = \frac{4!}{2!2!} = 6$

Number of ways to select one female out of 3 = ${}^3C_1 = \frac{3!}{1!(3-1)!} = \frac{3!}{1!2!} = 3$

By theorem 1.1:

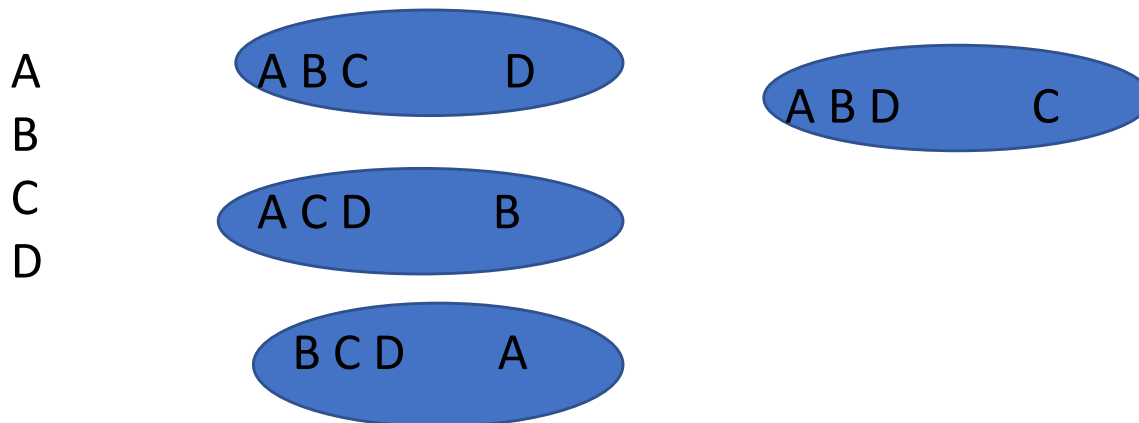
Total number of ways to select the scientists = **$6*3=18$**

Set Partition

A combination of r objects selected from a set of n distinct objects may be considered partition of n objects into two subsets containing respectively, the r objects that are selected and the $(n-r)$ objects that are left.

More general, problem of partitioning a set of n distinct objects into k subsets, which requires that each of n objects must belong to one and only one of the subsets.

The order of the subsets within a subset is of no importance.



Theorem 1.8

The number of ways in which a set of n distinct objects can be partitioned into k subsets with n_1 objects in the first subset, n_2 objects in the second subset, ..., and n_k objects in the k^{th} subset is

$$\binom{n}{n_1, n_2, \dots, n_k} = \frac{n!}{n_1! n_2! \dots n_k!}$$

Example 1.17:

How many ways can a set of seven objects are partitioned into three subsets containing respectively four, two and one of the objects?

$$\frac{7!}{4! 2! 1!} = \frac{7 * 6 * 5 * 4!}{(4!)(2)(1)} = 105$$

Example 1.18:

In how many ways can seven Businessmen attending a convention be assigned to one triple and two double hotel rooms?

$$\frac{7!}{3! 2! 2!} = \frac{7 * 6 * 5 * 4 * 3!}{3! * 2 * 2} = 210$$

Example 1.19:

A group of 100 students are planning to go on a field trip. There are four vehicles and the number of seats are as follows:

52 seat bus, 26 seat mini bus, 10 seat van and 12 seat van

How many different ways are there to divide 100 students into above four vehicles?

Example 1.20:

A passcode of 4 digits is needed to access an account. Calculate the total number of different passcodes that can be made if

- 1) All numbers (0-9) and all letters A-Z, uppercase or lowercase can be used and repetitions are allowed.
- 2) All numbers (0-9) and all letters A-Z, uppercase or lowercase can be used and repetitions are not allowed
- 3) All numbers (0-9) and all letters A-Z, uppercase or lowercase can be used and repetitions are not allowed and first digit cannot be 0.

Practice Question:

How many possible results are there for a draw of “Lagna Wasana?”
(The machine contains 61 balls:1-61 and 12 zodiac signs)

